

Requirements

Bug Report Classification Tool — Dependencies and Environment

Python Version

Python 3.9 or higher is required.

Required Packages

Package	Version	Purpose
scikit-learn	>=1.3	Machine learning classifiers, TF-IDF, metrics
pandas	>=2.0	Data loading and manipulation
numpy	>=1.24	Numerical operations
scipy	>=1.10	Wilcoxon signed-rank statistical test
matplotlib	>=3.7	Box plot figure generation
xgboost	>=1.7	XGBoost gradient boosted classifier
nltk	>=3.8	Stopword removal for text preprocessing

Installation

All dependencies are listed in `requirements.txt` at the root of the repository. To install:

```
pip install -r requirements.txt
```

If using a virtual environment (recommended):

```
python -m venv venv
```

```
source venv/bin/activate # macOS/Linux
```

```
pip install -r requirements.txt
```

NLTK Data

The script automatically downloads the required NLTK stopwords corpus on first run. An internet connection is needed for the initial run only.

Operating System

Tested on macOS 14 (Apple Silicon) and Ubuntu 22.04. Should work on any OS with Python 3.9+.